

Shubham Kumar Sharma

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Summary

Software Engineer specializing in **applied AI** and **full-stack architecture**. Engineered a production **RAG pipeline (AWS Bedrock)**, an automated mobile categorizer using **LLMs** and **vector search (React Native)**, and a live **DSA visualizer** handling complex **60 FPS** state management. **Top 1,500 of 50,000+ AWS AI & ML Scholars**.

Technical Skills

Languages: TypeScript, Python, Java, SQL, C	Databases: PostgreSQL, Supabase, Aurora Serverless
Backend: Node.js, Spring Boot, Prisma, Docker	Cloud & Infra: AWS, Terraform, IBM Cloud
Frontend & Mobile: React, React Native, Tailwind CSS, Vite	AI & Concepts: RAG, LLMs, Vector Search, Microservices, REST

Experience

IBM SkillsBuild Intern — AI & Cloud Technologies Edunet Foundation · Remote	July 2025 – Aug 2025
- Trained and deployed a Snap Random Forest classifier on IBM Watsonx.ai — 95.8% accuracy, outperforming logistic regression baseline by ~12pp on a government benefit applicant dataset. - Designed a decision support tool cutting manual applicant review time by 70% — removing the human classification bottleneck from a government benefit distribution pipeline.	

Projects

Intelligent Document Querying System (AWS Bedrock · RAG · Terraform · Production)	GitHub
Stack: AWS Bedrock, Terraform, Aurora Serverless PostgreSQL, S3, Python, Streamlit. - Built an end-to-end RAG pipeline — AWS Bedrock Knowledge Bases, Aurora Serverless PostgreSQL, S3 — delivering sub-second natural language search over multi-page technical PDFs. - Provisioned full AWS infrastructure via Terraform across two modular stacks (VPC, NAT, Aurora Serverless, S3, Bedrock KB, IAM) — reproducible end-to-end from a single terraform apply. - Implemented prompt validation via a Bedrock model pre-check — filtering irrelevant queries before KB invocation — and tuned temperature and top_p to balance retrieval precision with response diversity.	
Data Structures & Algorithms Visualizer (Live · 60 FPS · Full-Stack · 20+ Algorithms)	Live GitHub
Stack: React, TypeScript, Node.js, Express, Prisma, PostgreSQL, Zustand, Tailwind CSS, SVG, Canvas. - Built a custom Animation Engine separating algorithmic logic from UI rendering — stable 60 FPS on O(n²) sorts across 100+ element arrays, zero frame drops. - Replaced React state with Zustand — 30% fewer re-renders on worst-case O(n log n) sorts, verified via React DevTools Profiler; rendered 6+ data structures with live complexity overlays at every step. - Architected a full-stack system with Node.js + Express + Prisma + PostgreSQL backend, Jest test suite (health, catalog, auth flows), and GitHub Actions CI/CD — supporting 20+ algorithms across Arrays, Trees, Graphs (Dijkstra, A*, Kruskal, Prim), and Sorting.	
LaterLens (AI Screenshot Organizer · React Native · Vector Search)	GitHub
Stack: React Native, TypeScript, Supabase (pgvector), Python, OCR, LLM Embeddings. - Engineered an automated screenshot categorization engine, achieving 95% classification accuracy, by integrating cloud-based OCR with a custom LLM text-analysis pipeline. - Delivered a sub-second semantic search experience across thousands of images, reducing user retrieval time by 80%, by implementing a vector database architecture via pgvector and text embeddings. - Built a responsive cross-platform mobile interface, maintaining stable 60 FPS scrolling through heavy image grids, by utilizing React Native and Zustand for optimized local state management.	

Education

Engineering College, Ajmer — B.Tech in Information Technology CGPA: 9.06/10.0	Sept 2023 – Sept 2027
AWS AI & ML Scholars — Future AWS AI Engineer Nanodegree Udacity & AWS	Aug – Nov 2025
Selected: Top 1,500 of 50,000+ global applicants · ML, GenAI, SageMaker, Bedrock, Terraform	

Certifications

AWS ML Foundations — AWS (2025)	Intro to GenAI with AWS — Udacity (2025)	Getting Started with AI — IBM (2025)
Python Essentials 1 & 2 — Cisco (2025)		